2016 F=ma Exam Review

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Problem 1

A concise way of thinking:

$$\frac{\omega_L}{\omega_R} = \frac{\text{circle}_L/\text{time}}{\text{circle}_R/\text{time}} = \frac{2\pi R}{2\pi(R+1.74)}.$$

ω around the curve is the same for both, so v/r is the same for both, so

$$\frac{r\,\omega_{\text{left}}}{R} = \frac{r\,\omega_{\text{right}}}{R+1.74},$$

then you can find the ratio of ω_{left} to ω_{right}.

Problem 3

Energy conservation is best and concise: $E_i + W_{\text{ext}} = E_f$. Or use kinematics, and don't forget to *sum* the forces. The force up *also* has to overcome mg to accelerate it up.

Problem 4

There's tangential and centripetal acceleration; the question is whether a_c increases linearly or nonlinearly. $a_c = v^2/r$ and v depends on a constant forward acceleration times t, so v^2 depends on t^2, so the total acceleration is

$$a = \sqrt{(g\sin\theta)^2 + a_c^2} = \sqrt{(g\sin\theta)^2 + Ct^4},$$

where C is some constant. $\sqrt{Ct^4}$ isn't linear. Learn to simplify expressions about general relationships by subbing in constants.

Problem 8

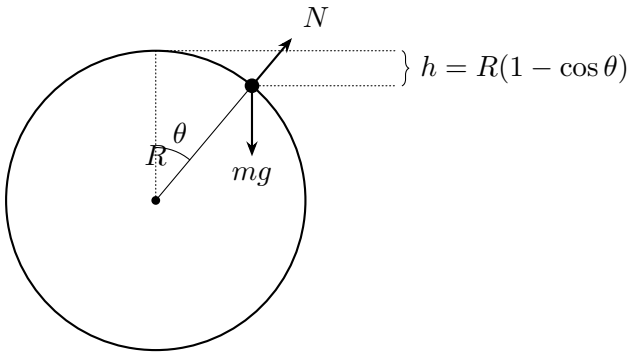
Conservation of angular momentum is independent of the nature of the central force. While it depends on r, it doesn't depend on the force, because a central force goes through the axis of rotation, providing no torque, meaning ΔL still $= 0$.

Problem 9

"Frictionless" → use energy conservation. The glass sphere does no work because it doesn't have any displacement and the normal force is always perpendicular to the bead's motion. Use circular motion:

$$mg\cos\theta - N = \frac{mv^2}{R}.$$

$N = 0$ when it loses contact. Some mathematical conveniences: $R = D/2$ (sub it in later). $h = R(1 - \cos\theta)$, you've seen that.



Problem 10

For tension T_2 to change, the string must change its length, and it can't do so because at that instant m_1 hasn't moved.

Problem 11

Write out what's given: $P \propto m$. The answer is in terms of v and m, so get rid of everything else by recognizing the relationship between m and A.

Problem 14

Use the $\ddot{x} = -\omega^2 x$ concept.

Problem 15

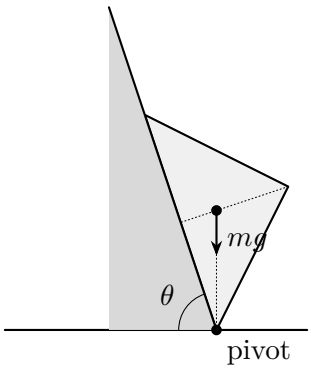
$\sin\theta \approx \theta$, use the same SHM concept. See that it's the same restoring force applied at the same distance → same frequency/period.

Problem 19

Write out equations with H, R, D, and relate them by plugging in. Energy is conserved, especially in these situations where something slides frictionlessly on a surface that doesn't move and exerts only an orthogonal force ($W = 0$).

Problem 20

No slipping condition: $\mu = \tan\theta$. When it reaches the angle where mg passes through the pivot point, it should have $\mu \geq \tan\theta$. θ can be found via triangle geometry (vertex to com : com to side midpoint $= 2:1$).



Problem 22

Energy (and horizontal momentum) of the system is conserved! The normal force from the ground does no work. Fact: a force does work only if the point of application moves in the direction of the force. Neither ball has horizontal velocity at the end (horizontal momentum is conserved and $= 0$). All the initial potential of the system gets transferred to kinetic, and the KE is concentrated in the top ball because the bottom ball has $v = 0$ and the rod is massless. The bottom ball cannot have v_x, otherwise the top mass must have the same v_x (they're horizontal, incompressible rod), but that would violate momentum conservation.

Review

Problem 6

Fact: uniformly accelerated motion is just parabolic.

Problem 17

0 torque about the IAOR (instantaneous axis of rotation), or contact point with the ground, so $\Delta L = 0$:

$$mvR + I\omega = 0, \qquad mvR = -\tfrac{2}{3}mR^2\omega, \qquad |v| = \tfrac{2}{3}R\omega.$$

Problem 25

Uncertain quantities added together have errors adding in quadrature.

- A: 2
- B: $\sqrt{2^2 + 2^2} = \sqrt{8} = 2\sqrt{2}$
- C: $\sqrt{2}$